Brandon Ling-xuan (Wade-Giles: hsuan) Chen
Researcher, Independent
Student, Island School, Hong Kong
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Taiwan and Hong Kong

Education

International Baccalaureate: Diploma Program – Island School, approx. June 2028

- Higher Level: Mathematics Analysis and Approaches, Physics, Economics
- Standard Level: Chemistry, Chinese B, English Language and Literature

Publications

- Shadow-Edge Timing for Urban Positioning with Low-Earth-Orbit Satellites - Working Paper
  Brandon Ling-xuan Chen

- Lightweight-Learning Assisted Pseudorange Reweighting for Smartphone GNSS in Urban Canyons - Conference Peer Review
  Brandon Ling-xuan Chen, Derek Hoi Wah Ng, Li-Ta Hsu

- Altitude-Dependent Geolocation Effects on Typhoon-Relative FORMOSAT-7/COSMIC-2 Radio Occultation Sampling and Refractivity Compositing - Journal Revision Process
  Brandon Ling-xuan Chen

- Altitude-Dependent Geolocation Effects on Typhoon-Relative FORMOSAT-7/COSMIC-2 Radio Occultation Sampling - Accepted, TASTI 2026
  Brandon Ling-xuan Chen

- Uncertainty-Aware 3D-Mapping-Aided GNSS with Probabilistic Building Models- Journal Peer Review
  Brandon Ling-xuan Chen

- The Value of Information in Smartphone 3D-Mapping-Aided GNSS - Journal Peer Review
  Brandon Ling-xuan Chen

Experience and Appointments

- Sept. 2026-Present: Research Assistant, Department of Aeronautical and Aviation Engineering, Hong Kong Polytechnic University
  - Research Assistant at the Intelligent Positioning and Navigation Laboratory (IPNL). Mentored, supervised, recruited and appointed by Professor Li-Ta Hsu
  - Supervised by Professor Li-Ta Hsu
  - Promoted from Independent Researcher by Prof. Li-Ta Hsu, Independent Researcher → Research Assistant
  - Work focuses on: Positioning, Navigation and Timing (PNT); Urban GNSS; AI/ML GNSS; 3D-Mapping-Aided GNSS; Smartphone and Wearables GNSS; Low-Earth Orbit (LEO) Positioning

- Apr 2026-Sept. 2026: Independent Researcher, Department of Aeronautical and Aviation Engineering, Hong Kong Polytechnic University

- Apr. 2026-Sept. 2026: Hardware and Software Engineer, Industrial Centre, Hong Kong

Polytechnic University
Mechanical and Electronic Engineer working in CAD under the Stairio R&D Project.
Appointed and supervised by Dr Max Lee Jwo-lem and Professor Li-Ta Hsu
Department of Mechanical Engineering and Department of Aeronautical and Aviation Engineering

- 2025-2026: Team Lead, Island School Typhoon Engineers, MIT EWB
  Team Lead within MIT'S EWB Competition for Science and Engineering
  Second Place Team in HK, Second in HK as an individual (90.32%)

- 2024-Present: Tutor in STEM, Self-Employment
  Tutored 15+ students in science and mathematics

- 2023-Present: Assistant Coach, Ortega Tennis Ltd.
  3+ years of Tennis Coaching
- 2023-Present: Musician, Self-Employment
  Freelance Musician with experience in both bands and as individual

Honors

Second Place in Hong Kong, Individual – 90.32%, Massachusetts Institute of Technology

Second Place in Hong Kong, Team, Massachusetts Institute of Technology

Certificate of Science Commendation, Island School

Spirit Award for Dedication to Extracurricular Activities, Island School

Professional and Academic Service
Member of the Institute of Navigation (2026-Present)
Student Assistant at the International Conference on Aerospace System Science Engineering (ICASSE) 2026
Editor-in-Chief, Island School Science Journal
STEM Committee Member, Island School
Head of Charity, Island School
Founder and President of Physics and Astrophysics Society, Island School
Teaching Assistant – Space Science, Island School Physics Department
Syllabus Support – Mathematics MYP + IGCSE, Island School Mathematics Department